

Fundamentals of Mathematics (25VBT0C102)



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Module: 02

Eigenvalues, Eigenvectors & Vector Algebra

Module Information

Field	Details
Course Title	Fundamentals of Mathematics
Course Code	25VBT0C102
Module Number	Module 2 of 5
Module Title	Eigenvalues, Eigenvectors, and Vector Algebra

Module Overview

Module 2 extends the foundation built in Module 1 into two rich and deeply connected branches of mathematics: eigenvalue theory and vector algebra. While the matrix operations of Module 1 described how to manipulate systems of equations, this module asks a deeper question — what are the intrinsic, invariant properties of a matrix itself? The answer lies in eigenvalues and eigenvectors.

Eigenvalues and eigenvectors are among the most profound and far-reaching concepts in all of mathematics. Every linear transformation — whether it rotates, scales, shears, or reflects — has special directions associated with it: directions along which the transformation acts purely as a scaling with no rotation. These directions are the eigenvectors, and the scaling factors are the eigenvalues. Understanding them reveals the "geometry" hidden inside the algebra of a matrix.

The module begins with the characteristic equation, which provides a systematic method to extract the eigenvalues of any square matrix. From eigenvalues, eigenvectors are then computed by solving homogeneous linear systems — directly applying the Gauss Elimination skills from Module 1. You will then study the important properties that eigenvalues satisfy, including the trace-sum and determinant-product relations.

Rayleigh's Power Method introduces a numerical approach for finding the dominant eigenvalue of a matrix — the one with the largest absolute value. This iterative technique is particularly valuable for large matrices where exact computation of all eigenvalues is impractical, and it underlies algorithms such as Google's original PageRank, used to rank web pages.

Diagonalization is the process of decomposing a matrix A into the form $A = PDP^{-1}$, where D is a diagonal matrix of eigenvalues and P is a matrix of eigenvectors. A diagonalised matrix is enormously easier to work with — computing its powers, exponential, and functions becomes trivial. This is exploited in physics, engineering, and data science.

The second half of the module introduces vector algebra — the mathematics of directed quantities. Vectors are essential for describing physical quantities that have both magnitude and direction: velocity, force, displacement, electric field. You will study the fundamental operations on vectors: addition, scalar multiplication, dot product, and cross product, each with clear geometric interpretation.

The dot product (scalar product) gives a measure of "how parallel" two vectors are and enables the computation of projections. The cross product (vector product) gives a vector perpendicular to two given vectors and enables the computation of areas and the torque in mechanics. The scalar triple product extends this to volumes.

Throughout both halves of the module, the emphasis is on geometric understanding backed by rigorous algebraic computation, so that you can apply these tools confidently in physics, engineering, computer graphics, and data analysis.

Learning Objectives

By the end of this module, you will be able to:

- Understand the definition of eigenvalues and eigenvectors and derive the characteristic equation of a square matrix.
- Apply the characteristic polynomial method to compute eigenvalues and corresponding eigenvectors for 2×2 and 3×3 matrices.
- Understand the diagonalization process and apply it to express a matrix in terms of its eigenvalues and eigenvectors.
- Understand and apply Rayleigh's Power Method to find the dominant eigenvalue iteratively.
- Apply vector operations — dot product, cross product, and scalar triple product — to solve geometric problems involving direction, angle, area, and volume.

Learning Outcomes

After completing this module, you will be able to:

- Compute eigenvalues by solving the characteristic equation $\det(A - \lambda I) = 0$ for 2×2 and 3×3 matrices.
- Find eigenvectors for each eigenvalue and verify them by substitution into the eigenvalue equation.
- Diagonalize a matrix $A = PDP^{-1}$ when it possesses a complete set of linearly independent eigenvectors.
- Perform Rayleigh's Power Method iterations to approximate the dominant eigenvalue of a real matrix.
- Compute dot products, cross products, projections, areas of parallelograms, and volumes of parallelepipeds using vector methods.

Table of Topics

Sub-Unit	Topic	Description
2.1	Eigenvalues — Definition and Characteristic Equation	Eigenvalue problem, $\det(A-\lambda I)=0$, characteristic polynomial
2.2	Computing Eigenvectors	Homogeneous system $(A-\lambda I)X=0$, null space, eigenvector basis
2.3	Properties of Eigenvalues	Trace-sum, det-product, symmetric matrix properties, Cayley-Hamilton
2.4	Rayleigh's Power Method	Iterative dominant eigenvalue, normalisation, convergence
2.5	Diagonalization of Matrices	Modal matrix, $A = PDP^{-1}$, conditions for diagonalizability
2.6	Linear Transformations and Matrix Reduction	Transformation as matrix, kernel, image, rank-nullity theorem
2.7	Introduction to Vectors — Types and Concepts	Scalar vs vector, position/unit/zero vectors, notation in 3D
2.8	Magnitude and Direction of Vectors	Modulus, direction cosines, direction ratios, unit vector
2.9	Vector Addition and Scalar Multiplication	Triangle and parallelogram laws, properties, linear combination
2.10	Dot Product and Geometric Meaning	$a \cdot b = a b \cos\theta$, orthogonality, angle between vectors
2.11	Projection of Vectors	Scalar and vector projection, component form
2.12	Cross Product of Vectors	$a \times b = a b \sin\theta \hat{n}$, determinant form, properties, torque
2.13	Scalar Triple Product, Area and Volume	$[a \ b \ c] = a \cdot (b \times c)$, volume of parallelepiped, coplanarity

2.1 Eigenvalues — Definition and the Characteristic Equation

Imagine applying a transformation to a vector: in general, the direction of the vector changes. But for certain special vectors — the eigenvectors — the transformation merely stretches or compresses them along the same direction, without any rotation. The factor by which the vector is scaled is called the eigenvalue. This concept is arguably the most important in all of applied linear algebra.

Formally, let A be an $n \times n$ square matrix. A scalar λ (lambda) is called an eigenvalue of A if there exists a non-zero vector X such that:

$$\text{Eigenvalue equation: } A \cdot X = \lambda \cdot X$$

The non-zero vector X satisfying this equation is called an eigenvector corresponding to the eigenvalue λ . The vector X must be non-zero — the zero vector trivially satisfies $AX = \lambda X$ for any λ , so it is excluded by definition.

Rearranging the eigenvalue equation: $AX - \lambda X = 0$, which can be written as $(A - \lambda I)X = 0$. This is a homogeneous system of linear equations. For a non-trivial (non-zero) solution X to exist, the coefficient matrix $(A - \lambda I)$ must be singular — its determinant must be zero:

$$\text{Characteristic equation: } \det(A - \lambda I) = 0$$

This equation is called the characteristic equation of A . When expanded, it is a polynomial equation in λ of degree n — called the characteristic polynomial. The n roots of this polynomial (counting multiplicity, including complex roots if any) are the eigenvalues of A .

Setting Up the Characteristic Equation

For an $n \times n$ matrix, the characteristic equation is obtained by:

- Step 1: Form the matrix $(A - \lambda I)$ by subtracting λ from every diagonal element of A .
- Step 2: Compute the determinant $\det(A - \lambda I)$.
- Step 3: Set this determinant equal to zero and solve for λ .

Worked Example: Characteristic Equation of a 2×2 Matrix

Find the eigenvalues of the matrix:

$$A =$$

4	1
2	3

Step 1: Form $A - \lambda I$ by subtracting λ from the diagonal:

$$A - \lambda I =$$

$4 - \lambda$	1
2	$3 - \lambda$

Step 2: Set $\det(A - \lambda I) = 0$:

$$\begin{aligned} \det(A - \lambda I) &= (4 - \lambda)(3 - \lambda) - (1)(2) \\ &= 12 - 4\lambda - 3\lambda + \lambda^2 - 2 \\ &= \lambda^2 - 7\lambda + 10 = 0 \end{aligned}$$

Step 3: Solve the characteristic equation by factoring:

$$(\lambda - 5)(\lambda - 2) = 0 \Rightarrow \lambda_1 = 5, \lambda_2 = 2$$

The eigenvalues of A are $\lambda_1 = 5$ and $\lambda_2 = 2$.

Worked Example: Characteristic Equation of a 3×3 Matrix

Find the eigenvalues of the matrix:

$$\mathbf{A} =$$

2	0	1
0	3	0
1	0	2

Step 1: Form $\mathbf{A} - \lambda \mathbf{I}$:

$$\mathbf{A} - \lambda \mathbf{I} =$$

$2-\lambda$	0	1
0	$3-\lambda$	0
1	0	$2-\lambda$

Step 2: Expand $\det(\mathbf{A} - \lambda \mathbf{I})$ along Row 2 (most zeros — simplifies calculation):

$$\det = (3-\lambda) \cdot \det([[2-\lambda, 1], [1, 2-\lambda]])$$

$$= (3-\lambda) \cdot [(2-\lambda)^2 - 1]$$

$$= (3-\lambda)(4 - 4\lambda + \lambda^2 - 1)$$

$$= (3-\lambda)(\lambda^2 - 4\lambda + 3)$$

$$= (3-\lambda)(\lambda-1)(\lambda-3)$$

Step 3: Set equal to zero: $(3-\lambda)(\lambda-1)(\lambda-3) = 0$

$$\lambda_1 = 1, \quad \lambda_2 = 3 \quad (\text{with multiplicity 2 since } (3-\lambda)(\lambda-3) = -(\lambda-3)^2)$$

Eigenvalues: $\lambda_1 = 1$ and $\lambda_2 = \lambda_3 = 3$.

💡 Did You Know?

Eigenvalues and eigenvectors are the mathematical engine behind Google's PageRank algorithm, which ranked web pages when Google was founded in 1998. The web is modelled as a matrix (the link matrix), and the dominant eigenvector of that matrix gives the relative importance of every web page. India's National Internet Exchange (NIXI), which manages the .in domain infrastructure, relies on similar graph-theoretic and linear-algebraic analysis for routing traffic. The concept of "dominant eigenvector" you will study in Rayleigh's Power Method is precisely what PageRank computes.

2.2 Computing Eigenvectors

Once the eigenvalues of a matrix are known, finding the corresponding eigenvectors is straightforward: for each eigenvalue λ , solve the homogeneous system $(A - \lambda I)X = 0$. Since the determinant of $(A - \lambda I)$ is zero for an eigenvalue, this system is guaranteed to have infinitely many non-trivial solutions. The set of all solutions forms the eigenspace of λ .

The eigenvectors are the non-zero vectors in this eigenspace. In practice, we use Gauss Elimination to reduce $(A - \lambda I)$ to Row Echelon Form and then express the solution in terms of free variables. The choice of free variable values determines the particular eigenvector (any scalar multiple of an eigenvector is also an eigenvector).

Worked Example: Eigenvectors for Each Eigenvalue — 2×2

Find the eigenvectors for A with eigenvalues $\lambda_1 = 5$ and $\lambda_2 = 2$:

$$A =$$

4	1
2	3

Step 1: For $\lambda_1 = 5$: Form $(A - 5I)$ and solve $(A - 5I)X = 0$:

$$A - 5I =$$

-1	1
2	-2

Augmented matrix $\rightarrow R_2 \rightarrow R_2 + 2R_1$:

RREF:

-1	1		0
0	0		0

From Row 1: $-x_1 + x_2 = 0 \Rightarrow x_1 = x_2$. Let $x_2 = t$ (free).

Eigenvector for $\lambda_1 = 5$: $X_1 = t[1; 1]$, or choosing $t = 1$: $X_1 = [1; 1]$

Step 2: Verify: $AX_1 = [[4,1],[2,3]][1;1] = [5; 5] = 5 \cdot [1;1] = 5X_1 \quad \checkmark$

Step 3: For $\lambda_2 = 2$: Form $(A - 2I)$:

$$A - 2I =$$

2	1
---	---

2	1
---	---

$R_2 \rightarrow R_2 - R_1$: Row 2 becomes $[0, 0 \mid 0]$.

From Row 1: $2x_1 + x_2 = 0 \Rightarrow x_2 = -2x_1$. Let $x_1 = t$ (free).

Eigenvector for $\lambda_2 = 2$: $X_2 = t[1; -2]$, or choosing $t = 1$: $X_2 = [1; -2]$

Step 4: Verify: $AX_2 = [[4,1],[2,3]][1;-2] = [4-2; 2-6] = [2;-4] = 2 \cdot [1;-2] = 2X_2 \quad \checkmark$

Worked Example: Eigenvectors for a 3×3 Matrix

Find the eigenvectors of A for each eigenvalue, given:

$A =$

2	0	1
0	3	0
1	0	2

Eigenvalues (from Sub-unit 2.1): $\lambda_1 = 1$, $\lambda_2 = \lambda_3 = 3$.

Step 1: For $\lambda_1 = 1$: Form $A - I$ and row-reduce:

$A - I =$

1	0	1
0	2	0
1	0	1

$R_3 \rightarrow R_3 - R_1$:

1	0	1
0	2	0
0	0	0

From Row 2: $2x_2 = 0 \Rightarrow x_2 = 0$.

From Row 1: $x_1 + x_3 = 0 \Rightarrow x_1 = -x_3$. Let $x_3 = t$.

Eigenvector for $\lambda_1 = 1$: $X_1 = [-1; 0; 1]$

Step 2: For $\lambda_2 = 3$: Form $A - 3I$ and row-reduce:

$$A - 3I =$$

-1	0	1
0	0	0
1	0	-1

$R_3 \rightarrow R_3 + R_1$: Row 3 becomes $[0,0,0]$. System reduces to: $-x_1 + x_3 = 0 \Rightarrow x_1 = x_3$.

x_2 is free (no constraint from non-zero rows). Let $x_1 = x_3 = s$, $x_2 = t$.

Two independent eigenvectors: $X_2 = [1; 0; 1]$ and $X_3 = [0; 1; 0]$.

(The eigenspace for $\lambda = 3$ is 2-dimensional — this matrix is diagonalizable.)

🔍 Analysing Misconceptions: Eigenvectors Are Unique

✗ MYTH: Each eigenvalue corresponds to exactly one eigenvector.

✓ REALITY: Eigenvectors are never unique. Any non-zero scalar multiple of an eigenvector is also an eigenvector for the same eigenvalue. Moreover, when an eigenvalue has algebraic multiplicity > 1 (a repeated eigenvalue), there may be multiple linearly independent eigenvectors spanning a multi-dimensional eigenspace. A matrix is diagonalizable precisely when the total number of linearly independent eigenvectors equals the matrix dimension.

2.3 Properties of Eigenvalues

Eigenvalues satisfy several remarkable algebraic relationships. These properties are not just theoretical curiosities — they provide powerful shortcuts for verification, and they underlie deep connections between the geometry of transformations and the algebra of determinants and traces.

Fundamental Properties

- The sum of all eigenvalues equals the trace of the matrix. The trace $\text{tr}(A)$ is the sum of the diagonal elements: $\lambda_1 + \lambda_2 + \dots + \lambda_n = \text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn}$.
- The product of all eigenvalues equals the determinant of the matrix: $\lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n = \det(A)$.
- A matrix A is singular ($\det(A) = 0$) if and only if at least one eigenvalue is zero.
- If λ is an eigenvalue of A , then λ^k is an eigenvalue of A^k (for any positive integer k), with the same eigenvector X .
- If λ is an eigenvalue of A and A is invertible, then $1/\lambda$ is an eigenvalue of A^{-1} .
- The eigenvalues of a triangular matrix (upper or lower) are exactly the diagonal elements.
- Eigenvalues of a real symmetric matrix are always real (never complex).

- Eigenvectors of a symmetric matrix corresponding to distinct eigenvalues are always orthogonal.

Cayley-Hamilton Theorem

The Cayley-Hamilton Theorem states that every square matrix satisfies its own characteristic equation. If the characteristic polynomial of A is $p(\lambda) = \lambda^n + c_1\lambda^{n-1} + \dots + c_n$, then substituting the matrix A for λ gives the zero matrix:

$$\text{Cayley-Hamilton Theorem: } p(A) = A^n + c_1A^{n-1} + \dots + c_nI = O$$

This theorem has a beautiful practical application: it allows you to express any higher power of A (such as A^3 , A^4 , or even A^{-1}) as a linear combination of lower powers of A , using the characteristic equation to reduce powers.

Worked Example: Applying Cayley-Hamilton Theorem

For $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$, verify Cayley-Hamilton and use it to find A^3 .

Step 1: Find the characteristic polynomial:

$$\det(A - \lambda I) = (1 - \lambda)(2 - \lambda) - 6 = \lambda^2 - 3\lambda + 2 - 6 = \lambda^2 - 3\lambda - 4 = 0$$

Step 2: By Cayley-Hamilton: $A^2 - 3A - 4I = O \implies A^2 = 3A + 4I$

$$\text{Compute } A^2: \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 9 & 10 \end{bmatrix}$$

$$\begin{aligned} \text{Verify: } 3A + 4I &= 3\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} + 4\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 6 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 9 & 10 \end{bmatrix} \\ &= A^2 \quad \checkmark \end{aligned}$$

Step 3: Find $A^3 = A \cdot A^2 = A(3A + 4I) = 3A^2 + 4A = 3(3A + 4I) + 4A = 9A + 12I + 4A = 13A + 12I$

$$\begin{aligned} A^3 &= 13\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} + 12\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 13 & 26 \\ 39 & 26 \end{bmatrix} + \begin{bmatrix} 12 & 0 \\ 0 & 12 \end{bmatrix} = \\ &= \begin{bmatrix} 25 & 26 \\ 39 & 38 \end{bmatrix} \end{aligned}$$

 Let's Check

Q.No	Type	Question
1	MCQ	For a 2×2 matrix with eigenvalues 3 and 5, what is the value of its determinant?
2	MCQ	The eigenvalues of an upper triangular matrix are found by:
3	Fill in Blank	The sum of all eigenvalues of a matrix equals the _____ of the matrix.
4	Fill in Blank	By the Cayley-Hamilton theorem, every square matrix satisfies its own _____ equation.
5	True/False	If A has eigenvalue λ , then A^2 has eigenvalue λ^2 with the same eigenvector.
6	True/False	A matrix is singular if and only if zero is one of its eigenvalues.

Answers:

Q.No	Answer
1	c) 15 — $\det(A) = \text{product of eigenvalues} = 3 \times 5 = 15$.
2	a) Reading the diagonal entries directly — they are the eigenvalues.
3	trace (sum of diagonal elements)
4	characteristic
5	True — if $AX = \lambda X$, then $A^2X = A(AX) = A(\lambda X) = \lambda(AX) = \lambda^2X$.
6	True — $\det(A) = \text{product of eigenvalues}$; $\det = 0$ iff at least one eigenvalue = 0.

2.4 Rayleigh's Power Method

While the characteristic equation gives the exact eigenvalues of a matrix, it becomes increasingly difficult to solve as matrix size grows. For a 10×10 matrix, the characteristic polynomial is of degree 10 — finding its roots analytically is generally impossible. Rayleigh's Power Method (also called the Power Iteration method) provides an elegant numerical approach: by repeatedly multiplying a vector by the matrix and normalising, the iteration converges to the dominant eigenvector and dominant eigenvalue.

The dominant eigenvalue is the one with the largest absolute value. In many applications — principal component analysis, structural vibration modes, and PageRank — it is precisely this dominant eigenvalue that matters most. Power Method is computationally cheap and converges reliably for most practical matrices.

The Power Method Algorithm

Given an $n \times n$ matrix A :

- Step 1: Choose an initial non-zero vector X_0 (often the vector of all ones: $X_0 = [1; 1; \dots; 1]$).
- Step 2: Compute $Y_1 = A \cdot X_0$.
- Step 3: Identify $m_1 =$ the element of Y_1 with the largest absolute value (this is the Rayleigh quotient approximation).
- Step 4: Normalise: $X_1 = Y_1 / m_1$.
- Step 5: Repeat with $X_1 \rightarrow Y_2 = A \cdot X_1 \rightarrow m_2 = \max \text{ element} \rightarrow X_2 = Y_2 / m_2$.
- Step 6: Continue until m_k converges ($|m_k - m_{k-1}| < \epsilon$ for a chosen tolerance ϵ).

At convergence: $m_k \approx$ dominant eigenvalue λ_1 , and $X_k \approx$ corresponding eigenvector.

Worked Example: Rayleigh's Power Method — 4 Iterations

Use the Power Method to find the dominant eigenvalue of:

$A =$

2	1
1	3

Start with $X_0 = [1; 1]$.

Step 1: Iteration 1:

$$Y_1 = A \cdot X_0 = [[2, 1], [1, 3]] \cdot [1; 1] = [2+1; 1+3] = [3; 4]$$

$m_1 = 4$ (largest absolute value in Y_1)

$$X_1 = Y_1/m_1 = [3/4; 1] = [0.75; 1]$$

Step 2: Iteration 2:

$$Y_2 = A \cdot X_1 = [[2, 1], [1, 3]] \cdot [0.75; 1] = [1.5+1; 0.75+3] = [2.5; 3.75]$$

$$m_2 = 3.75, \quad X_2 = [2.5/3.75; 1] = [0.6667; 1]$$

Step 3: Iteration 3:

$$Y_3 = [[2, 1], [1, 3]] \cdot [0.6667; 1] = [1.3334+1; 0.6667+3] = [2.3334; 3.6667]$$

$$m_3 = 3.6667, \quad X_3 = [2.3334/3.6667; 1] = [0.6364; 1]$$

Step 4: Iteration 4:

$$Y_4 = [[2, 1], [1, 3]] \cdot [0.6364; 1] = [1.2728+1; 0.6364+3] = [2.2728; 3.6364]$$

$$m_4 = 3.6364, \quad X_4 = [0.625; 1]$$

Converging: $m \rightarrow 3.618...$ (Exact: dominant eigenvalue $= (5 + \sqrt{5})/2 \approx 3.618$)

Eigenvector direction: $X \approx [0.618; 1]$, proportional to $[(\sqrt{5}-1)/2; 1]$.

(Note: Exact eigenvalues of A are $\lambda = (5 \pm \sqrt{5})/2 \approx 3.618$ and 1.382 .)

★ **Note:** Power Method converges to the dominant eigenvalue assuming the dominant eigenvalue is strictly larger in magnitude than all others. Convergence is slower when two eigenvalues are close in magnitude. For the subdominant eigenvalue, deflation techniques are used — subtract the dominant contribution and apply Power Method again.

2.5 Diagonalization of Matrices

Diagonalization is one of the most powerful tools in linear algebra. The idea is to find a change of coordinates in which a matrix acts simply — as a pure scaling in each coordinate direction, with no mixing between coordinates. In such a coordinate system, the matrix is represented by a diagonal matrix D , and all computations become trivially simple.

A square matrix A of order n is said to be diagonalizable if there exists an invertible matrix P such that:

$$\textbf{Diagonalization: } A = P \cdot D \cdot P^{-1} \quad \text{or equivalently} \quad D = P^{-1} \cdot A \cdot P$$

Here, D is a diagonal matrix whose diagonal entries are the eigenvalues of A , and P is a matrix whose columns are the corresponding eigenvectors. The matrix P is called the modal matrix or eigenvector matrix.

Conditions for Diagonalizability

An $n \times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors. In particular:

- If A has n distinct eigenvalues, it is guaranteed to be diagonalizable (eigenvectors of distinct eigenvalues are always linearly independent).

- If A has repeated eigenvalues, it may or may not be diagonalizable — it depends on whether each repeated eigenvalue has a full set of independent eigenvectors (geometric multiplicity = algebraic multiplicity).
- All real symmetric matrices are diagonalizable (and their eigenvectors can be chosen to be orthogonal — orthogonal diagonalization).

Why is Diagonalization Useful?

Once $A = PDP^{-1}$, computing powers becomes trivial:

$$\text{Powers via diagonalization: } A^k = P \cdot D^k \cdot P^{-1}$$

And D^k is simply the diagonal matrix with entries $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$ — just raise each eigenvalue to the power k . Without diagonalization, computing A^{10} would require 9 matrix multiplications; with diagonalization, it requires only computing λ_i^{10} for each eigenvalue.

Worked Example: Diagonalizing a 2x2 Matrix

Diagonalize $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ (eigenvalues $\lambda_1=5, \lambda_2=2$; eigenvectors $X_1=\begin{bmatrix} 1 \\ 1 \end{bmatrix}, X_2=\begin{bmatrix} 1 \\ -2 \end{bmatrix}$).

Step 1: Form P using eigenvectors as columns (first column for $\lambda_1=5$, second for $\lambda_2=2$):

$$P =$$

1	1
1	-2

Step 2: Form D using eigenvalues on the diagonal (in same order as columns of P):

$$D =$$

5	0
---	---

0	2
---	---

Step 3: Compute P^{-1} . $\det(P) = (1)(-2) - (1)(1) = -3$.

$$\text{adj}(P) = \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix}. \quad P^{-1} = (1/-3) \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$

Step 4: Verify $A = PDP^{-1}$:

$$PD = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix}$$

$$(PD) \cdot P^{-1} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix} \cdot \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$$

$$= \begin{bmatrix} 10/3 + 2/3 & 5/3 - 2/3 \\ 10/3 - 4/3 & 5/3 + 4/3 \end{bmatrix}$$

$$= \begin{bmatrix} 12/3 & 3/3 \\ 6/3 & 9/3 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = A \quad \checkmark$$

2.6 Linear Transformations and Matrix Reduction

Every matrix can be interpreted as a linear transformation — a function that maps vectors to vectors, preserving the operations of addition and scalar multiplication. This geometric perspective reveals why eigenvalues and eigenvectors are so fundamental: they identify the "axes" of the transformation — directions that are scaled but not rotated.

Definition of a Linear Transformation

A function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation if it satisfies two conditions for all vectors u, v in $\mathbb{R}^n$ and all scalars c :

- Additivity: $T(u + v) = T(u) + T(v)$
- Homogeneity: $T(cu) = cT(u)$

Every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be represented by an $m \times n$ matrix A such that $T(X) = AX$. Conversely, every $m \times n$ matrix defines a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Kernel and Image

Two fundamental subspaces are associated with every linear transformation T represented by matrix A :

- Kernel (Null Space) of T : The set of all vectors X such that $AX = 0$. Its dimension is called the nullity of A , denoted $\text{null}(A)$. The kernel is the set of inputs that get "collapsed" to zero by the transformation.
- Image (Column Space) of T : The set of all possible outputs AX . Its dimension equals the rank of A , denoted $\text{rank}(A)$. The image is all possible outputs of the transformation.

Rank-Nullity Theorem

The Rank-Nullity Theorem connects these two fundamental dimensions. For a linear transformation represented by an $m \times n$ matrix A :

Rank-Nullity Theorem: $\text{rank}(A) + \text{nullity}(A) = n$ (number of columns / input dimension)

This elegant result says that the "information" carried by the transformation (its rank — how many independent output directions it creates) plus the "information lost" (its nullity — how many input dimensions are collapsed to zero) must always add up to the full input dimension n .

 Worked Example: Rank-Nullity Verification

Verify the Rank-Nullity theorem for the 3×4 matrix:

A =

1	2	0	3
0	1	1	2
1	3	1	5

Step 1: Row-reduce A to find rank:

$R_3 \rightarrow R_3 - R_1$:

1	2	0	3
0	1	1	2
0	1	1	2

$R_3 \rightarrow R_3 - R_2$:

REF:

1	2	0	3
0	1	1	2
0	0	0	0

Two non-zero rows $\Rightarrow \text{rank}(A) = 2$.

Step 2: By Rank-Nullity: $\text{nullity}(A) = n - \text{rank}(A) = 4 - 2 = 2$.

(There are 2 free variables in the null space, confirming nullity = 2.) ✓

2.7 Introduction to Vectors — Types and Concepts

We now transition to the second major topic of this module: vector algebra. While the first half dealt with the algebra of matrices and linear transformations, vector algebra is the language of geometry and physics. Vectors allow us to describe quantities that have both magnitude and direction — velocity, force, displacement, electric and magnetic fields — and to compute with them algebraically.

A scalar is a quantity described entirely by a single number (magnitude), such as temperature, mass, or time. A vector is a quantity described by both a magnitude and a direction, such as force, velocity, or displacement. In mathematics, we represent a vector as an ordered tuple of numbers — its components — which encode both magnitude and direction together.

Notation

Vectors in three-dimensional space are typically written as:

- In bold: $\mathbf{a}$, $\mathbf{F}$, $\mathbf{v}$ (common in printed text)
- With arrow: $\vec{a}$, $\vec{F}$, $\vec{v}$ or $\rightarrow\mathbf{a}$, $\rightarrow\mathbf{F}$ (common in handwriting)
- In component form: $\mathbf{a} = (a_1, a_2, a_3)$ or $\mathbf{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

Here, $\hat{i}$, $\hat{j}$, $\hat{k}$ are the standard unit vectors along the positive x, y, and z axes respectively. Every vector in 3D space can be expressed as a unique linear combination of these three unit vectors.

Types of Vectors

Type	Definition	Example / Notation
Zero Vector	A vector with magnitude 0 and no specific direction. All components are zero.	$0 = 0\hat{i} + 0\hat{j} + 0\hat{k} = (0, 0, 0)$
Unit Vector	A vector with magnitude exactly 1. Used to specify direction only.	$\hat{i} = (1,0,0), \hat{j} = (0,1,0), \hat{k} = (0,0,1)$
Position Vector	Vector from the origin O to a point P(x, y, z).	$OP^{\rightarrow} = x\hat{i} + y\hat{j} + z\hat{k}$
Equal Vectors	Two vectors with the same magnitude and direction, regardless of starting point.	$a^{\rightarrow} = b^{\rightarrow}$ if $ a^{\rightarrow} = b^{\rightarrow} $ and direction is same
Negative Vector	A vector with the same magnitude but opposite direction.	$-a^{\rightarrow}$ has same magnitude as $a^{\rightarrow}$

Collinear Vectors	Vectors that are parallel (same or opposite direction).	$\vec{a} = \lambda \vec{b}$ for some scalar λ
Coplanar Vectors	Three or more vectors lying in the same plane.	Scalar triple product $[a \ b \ c] = 0$
Free Vector	A vector that can be placed anywhere in space (position independent).	Most vectors in algebra are free vectors

2.8 Magnitude, Direction, and Unit Vectors

The magnitude (or modulus or length) of a vector is the measure of how large it is — the scalar that quantifies its size independent of direction. For a vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, the magnitude is given by the three-dimensional extension of the Pythagorean theorem:

$$\text{Magnitude of a vector: } |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

This formula emerges naturally from the distance formula in 3D space. It is the length of the vector when drawn as an arrow from the origin to the point (a_1, a_2, a_3) .

Unit Vector in the Direction of $\vec{a}$

A unit vector $\hat{a}$ (read "a-hat") is any vector with magnitude 1. Given any non-zero vector $\vec{a}$, we can construct a unit vector in the same direction by dividing by the magnitude:

$$\text{Unit vector: } \hat{a} = \vec{a} / |\vec{a}| = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) / \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Unit vectors are fundamental for specifying direction without any information about magnitude. Every direction in 3D space corresponds to a unique unit vector (on the unit sphere $|a| = 1$).

Direction Cosines and Direction Ratios

The direction of a vector in 3D space can be described by the angles it makes with the three coordinate axes. If $a = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ makes angles α, β, γ with the x, y, z axes respectively, then the direction cosines are:

Direction cosines: $\cos \alpha = a_1/|a|, \quad \cos \beta = a_2/|a|, \quad \cos \gamma = a_3/|a|$

These satisfy the identity: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

The quantities $a_1 : a_2 : a_3$ (the components without dividing by magnitude) are called the direction ratios. Any scalar multiple of the direction ratios represents the same direction. The direction cosines are the direction ratios of the unit vector.

Worked Example: Magnitude, Unit Vector and Direction Cosines

For the vector $a = 2\hat{i} - 3\hat{j} + 6\hat{k}$, find: (i) $|a|$, (ii) the unit vector $\hat{a}$, and (iii) direction cosines.

Step 1: Magnitude:

$$|a| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

Step 2: Unit vector:

$$\hat{a} = a / |a| = (1/7)(2\hat{i} - 3\hat{j} + 6\hat{k}) = (2/7)\hat{i} - (3/7)\hat{j} + (6/7)\hat{k}$$

Step 3: Direction cosines:

$$\cos \alpha = 2/7, \quad \cos \beta = -3/7, \quad \cos \gamma = 6/7$$

$$\text{Verify: } (2/7)^2 + (-3/7)^2 + (6/7)^2 = 4/49 + 9/49 + 36/49 = 49/49 = 1 \quad \checkmark$$

2.9 Vector Addition and Scalar Multiplication

Vector addition and scalar multiplication are the two foundational operations of vector algebra. They are geometrically intuitive — addition corresponds to placing vectors tip to tail, while scalar multiplication stretches or compresses a vector. Together, they give vectors the structure of a vector space, which is the abstract setting for all of linear algebra.

Vector Addition

The sum of two vectors $a = (a_1, a_2, a_3)$ and $b = (b_1, b_2, b_3)$ is computed component-wise:

$$\textbf{Vector addition: } a + b = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$$

Geometrically, vector addition follows the Triangle Law: place the tail of b at the tip of a ; the resultant $a + b$ is the vector from the tail of a to the tip of b . Equivalently, in the Parallelogram Law: the diagonal of the parallelogram formed by a and b (placed tail-to-tail) is $a + b$.

Properties of Vector Addition

- Commutativity: $a + b = b + a$
- Associativity: $(a + b) + c = a + (b + c)$
- Additive identity: $a + 0 = a$ (where 0 is the zero vector)
- Additive inverse: $a + (-a) = 0$

Scalar Multiplication

Multiplying a vector a by a scalar $m \in \mathbb{R}$ scales its magnitude by $|m|$ and preserves or reverses its direction depending on the sign of m :

$$\textbf{Scalar multiplication: } ma = (ma_1)\hat{i} + (ma_2)\hat{j} + (ma_3)\hat{k}$$

If $m > 0$, ma points in the same direction as a . If $m < 0$, ma points in the opposite direction. If $m = 0$, the result is the zero vector.

Linear Combination of Vectors

Any vector r in 3D space can be written as a linear combination of the three standard unit vectors: $r = r_1\hat{i} + r_2\hat{j} + r_3\hat{k}$. More generally, a vector v is a linear combination of vectors a, b, c if $v = \alpha a + \beta b + \gamma c$ for some scalars α, β, γ . The concept of linear combination from matrix algebra extends naturally to vector algebra.

Worked Example: Vector Addition and Subtraction

Given $a = 3\hat{i} - 2\hat{j} + \hat{k}$ and $b = \hat{i} + 4\hat{j} - 2\hat{k}$, compute: (i) $a + b$, (ii) $a - b$, (iii) $2a - 3b$.

Step 1: $a + b = (3+1)\hat{i} + (-2+4)\hat{j} + (1-2)\hat{k} = 4\hat{i} + 2\hat{j} - \hat{k}$

Step 2: $a - b = (3-1)\hat{i} + (-2-4)\hat{j} + (1+2)\hat{k} = 2\hat{i} - 6\hat{j} + 3\hat{k}$

Step 3: $2a = 6\hat{i} - 4\hat{j} + 2\hat{k}$ and $3b = 3\hat{i} + 12\hat{j} - 6\hat{k}$

$2a - 3b = (6-3)\hat{i} + (-4-12)\hat{j} + (2+6)\hat{k} = 3\hat{i} - 16\hat{j} + 8\hat{k}$

2.10 Dot Product and Its Geometric Meaning

The dot product (also called the scalar product or inner product) is one of the two fundamental ways to "multiply" two vectors. Unlike ordinary multiplication which always produces a result of the same type, the dot product of two vectors produces a scalar — a single number. This makes it a powerful tool for measuring the "alignment" between two vectors.

Definition of the Dot Product

For two vectors $a = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $b = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, the dot product is:

Algebraic definition: $a \cdot b = a_1b_1 + a_2b_2 + a_3b_3$

Geometric Definition and Angle Between Vectors

The dot product has a beautiful geometric interpretation. If θ is the angle between vectors a and b (where $0 \leq \theta \leq \pi$), then:

$$\text{Geometric definition: } a \cdot b = |a| \cdot |b| \cdot \cos \theta$$

This immediately gives us the angle between any two vectors:

$$\text{Angle formula: } \cos \theta = (a \cdot b) / (|a| \cdot |b|)$$

This formula is one of the most useful in all of geometry — it enables us to compute angles between lines, planes, and any geometric objects that can be described by direction vectors.

Properties of the Dot Product

- Commutativity: $a \cdot b = b \cdot a$
- Distributivity: $a \cdot (b + c) = a \cdot b + a \cdot c$
- Scalar association: $(ma) \cdot b = m(a \cdot b)$
- Self-dot product: $a \cdot a = |a|^2$ (the square of the magnitude)
- Orthogonality: $a \cdot b = 0$ if and only if $a \perp b$ (a and b are perpendicular)

Key Applications

- Orthogonality test: Two non-zero vectors are perpendicular if and only if their dot product is zero.
- Angle calculation: $\cos \theta = (a \cdot b) / (|a| \cdot |b|)$.
- Work done by force: $W = F \cdot d = |F||d|\cos\theta$ (the dot product gives scalar work).
- Sign interpretation: $a \cdot b > 0$ means angle $< 90^\circ$ (vectors lean toward each other); $a \cdot b < 0$ means angle $> 90^\circ$ (vectors lean away); $a \cdot b = 0$ means exactly perpendicular.

Worked Example: Dot Product — Angle Between Vectors

Find the dot product and the angle between $\mathbf{a} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\mathbf{b} = 2\hat{i} - \hat{j} + 2\hat{k}$.

Step 1: Compute the dot product:

$$\mathbf{a} \cdot \mathbf{b} = (1)(2) + (2)(-1) + (-2)(2) = 2 - 2 - 4 = -4$$

Step 2: Compute magnitudes:

$$|\mathbf{a}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

$$|\mathbf{b}| = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

Step 3: Find the angle:

$$\cos \theta = (\mathbf{a} \cdot \mathbf{b}) / (|\mathbf{a}||\mathbf{b}|) = -4 / (3 \cdot 3) = -4/9$$

$$\theta = \cos^{-1}(-4/9) \approx 116.4^\circ$$

Since $\mathbf{a} \cdot \mathbf{b} < 0$, the angle is obtuse ($> 90^\circ$). ✓

Let's Check

Q.No	Type	Question
1	MCQ	The dot product of two perpendicular vectors is:
2	MCQ	If $\mathbf{a} \cdot \mathbf{a} = 49$, then $ \mathbf{a} $ equals:
3	Fill in Blank	The formula for the angle θ between vectors $\mathbf{a}$ and $\mathbf{b}$ is: $\cos \theta = \underline{\hspace{2cm}}$.
4	Fill in Blank	The dot product of two vectors produces a _____ (scalar/vector).
5	True/False	If $\mathbf{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\mathbf{b} = \hat{i} + 2\hat{j}$, then $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

6	True/False	$a \cdot b = b \cdot a$ for any two vectors a and b (dot product is commutative).
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Answers:

Q.No	Answer
1	c) Zero — if $a \perp b$, then $\cos 90^\circ = 0$, so $a \cdot b = a b \cdot 0 = 0$.
2	b) 7 — since $a \cdot a = a ^2 = 49$, therefore $ a = 7$.
3	$(a \cdot b) / (a \cdot b)$
4	scalar
5	True — $a \cdot b = (2)(1) + (-1)(2) + (2)(0) = 2 - 2 + 0 = 0$. Zero dot product $\Rightarrow$ perpendicular.
6	True — $a \cdot b = a_1b_1 + a_2b_2 + a_3b_3 = b_1a_1 + b_2a_2 + b_3a_3 = b \cdot a$.

2.11 Projection of Vectors

The projection of one vector onto another is the shadow of the first vector cast along the direction of the second, when light is shining perpendicularly. This concept is central to decomposing forces in mechanics, computing components of velocities in particular directions, and forming the basis of least-squares fitting in statistics.

Scalar Projection (Component)

The scalar projection of vector a onto vector b is the signed length of the shadow of a along b . It is a scalar — it may be positive, negative, or zero depending on whether a points with, against, or perpendicular to b :

Scalar projection: $\text{Scalar projection of } a \text{ onto } b = (a \cdot b) / |b| = |a| \cos \theta$

Vector Projection

The vector projection of a onto b is the vector along b whose magnitude equals the scalar projection. It is obtained by multiplying the scalar projection by the unit vector in the direction of b :

Vector projection: $\text{proj}_b(a) = [(a \cdot b) / |b|^2] \cdot b = [(a \cdot b) / (b \cdot b)] \cdot b$

Decomposition of a Vector

Any vector a can be uniquely decomposed into two components:

- The component parallel to b (the projection): $a_{\parallel} = \text{proj}_b(a)$
- The component perpendicular to b (the rejection): $a_{\perp} = a - \text{proj}_b(a)$

This decomposition satisfies $a = a_{\parallel} + a_{\perp}$, where $a_{\parallel} \cdot a_{\perp} = 0$ (they are orthogonal).

 Worked Example: Scalar and Vector Projection

Find the scalar and vector projection of $\mathbf{a} = 3\hat{i} + 2\hat{j} - \hat{k}$ onto $\mathbf{b} = \hat{i} + \hat{j} + \hat{k}$.

Step 1: Compute $\mathbf{a} \cdot \mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = (3)(1) + (2)(1) + (-1)(1) = 3 + 2 - 1 = 4$$

Step 2: Compute $|\mathbf{b}|$:

$$|\mathbf{b}| = \sqrt{1 + 1 + 1} = \sqrt{3}$$

Step 3: Scalar projection of $\mathbf{a}$ onto $\mathbf{b}$:

$$(\mathbf{a} \cdot \mathbf{b})/|\mathbf{b}| = 4/\sqrt{3} = 4\sqrt{3}/3 \approx 2.309$$

Step 4: Vector projection of $\mathbf{a}$ onto $\mathbf{b}$:

$$\text{proj}_{\mathbf{b}}(\mathbf{a}) = [(\mathbf{a} \cdot \mathbf{b})/(\mathbf{b} \cdot \mathbf{b})] \cdot \mathbf{b} = [4/3](\hat{i} + \hat{j} + \hat{k}) = (4/3)\hat{i} + (4/3)\hat{j} + (4/3)\hat{k}$$

Step 5: Perpendicular component (rejection):

$$\mathbf{a}_{\perp} = \mathbf{a} - \text{proj}_{\mathbf{b}}(\mathbf{a}) = (3 - 4/3)\hat{i} + (2 - 4/3)\hat{j} + (-1 - 4/3)\hat{k} = (5/3)\hat{i} + (2/3)\hat{j} - (7/3)\hat{k}$$

$$\text{Verify orthogonality: } \mathbf{a}_{\perp} \cdot \mathbf{b} = (5/3)(1) + (2/3)(1) + (-7/3)(1) = 5/3 + 2/3 - 7/3 = 0$$

✓

2.12 Cross Product of Vectors

The cross product (also called the vector product) is the second fundamental way to "multiply" two vectors. Unlike the dot product which produces a scalar, the cross product produces a vector. This new vector is perpendicular to both original vectors — it points in the direction given by the right-hand rule. The cross product is the mathematical encoding of rotation, torque, and area in physics.

Definition of the Cross Product

For two vectors $\mathbf{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\mathbf{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, the cross product $\mathbf{a} \times \mathbf{b}$ is:

Cross product components: $\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\hat{i} - (a_1b_3 - a_3b_1)\hat{j} + (a_1b_2 - a_2b_1)\hat{k}$

A convenient way to remember this formula is through the formal determinant expansion:

$$\mathbf{a} \times \mathbf{b} = \det$$

$\hat{i}$	$\hat{j}$	$\hat{k}$
a_1	a_2	a_3
b_1	b_2	b_3

This 3×3 determinant is expanded along the first row, giving the component formula above.

Geometric Interpretation

The cross product $\mathbf{a} \times \mathbf{b}$ has a beautiful geometric meaning:

Magnitude of cross product: $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| \cdot |\mathbf{b}| \cdot \sin \theta$

where θ is the angle between a and b ($0 \leq \theta \leq \pi$). The direction of $a \times b$ is perpendicular to both a and b , following the right-hand rule: curl the fingers of your right hand from a toward b ; your thumb points in the direction of $a \times b$.

The magnitude $|a \times b|$ equals the area of the parallelogram formed by a and b . This is the most direct geometric application of the cross product.

Properties of the Cross Product

- Anti-commutativity: $a \times b = -(b \times a)$ — reversing the order reverses the direction.
- Parallel vectors: $a \times b = 0$ if and only if a and b are parallel ($\sin 0^\circ = 0$ or $\sin 180^\circ = 0$).
- Self-cross product: $a \times a = 0$ for any vector a .
- Distributivity: $a \times (b + c) = (a \times b) + (a \times c)$.
- Scalar association: $(ma) \times b = m(a \times b)$.
- Standard basis: $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (cyclic).

Worked Example: Cross Product Computation

Find $a \times b$ where $a = 2\hat{i} + 3\hat{j} - \hat{k}$ and $b = \hat{i} - 2\hat{j} + 4\hat{k}$.

Step 1: Set up the determinant:

$$a \times b = \det$$

$\hat{i}$	$\hat{j}$	$\hat{k}$
2	3	-1
1	-2	4

Step 2: Expand along Row 1:

$$\hat{i}\text{-component: } +\det([[3, -1], [-2, 4]]) = +(12 - 2) = +10$$

$$\hat{j}\text{-component: } -\det([[2, -1], [1, 4]]) = -(8 - (-1)) = -9$$

$\hat{k}$ -component: $+\det([[2,3],[1,-2]]) = +(-4 - 3) = -7$

Step 3: Result: $a \times b = 10\hat{i} - 9\hat{j} - 7\hat{k}$

Step 4: Verify perpendicularity with a:

$$(a \times b) \cdot a = (10)(2) + (-9)(3) + (-7)(-1) = 20 - 27 + 7 = 0 \quad \checkmark$$

Step 5: Verify perpendicularity with b:

$$(a \times b) \cdot b = (10)(1) + (-9)(-2) + (-7)(4) = 10 + 18 - 28 = 0 \quad \checkmark$$

Q Analysing Misconceptions: Cross Product is Commutative

✗ MYTH: Like the dot product, the cross product satisfies $a \times b = b \times a$.

✓ REALITY: The cross product is anti-commutative, not commutative: $a \times b = -(b \times a)$. Reversing the order of the cross product reverses the direction of the resulting vector (from the right-hand rule). This is why torque $r \times F \neq F \times r$ — the physical direction of rotation changes when the order is reversed. Always be careful about the order in cross products.

2.13 Scalar Triple Product, Area, and Volume

The scalar triple product combines both the cross product and the dot product to create a single scalar from three vectors. It has remarkable geometric significance: it directly computes the volume of a parallelepiped (the 3D generalisation of a parallelogram) formed by three vectors. This makes it an indispensable tool in geometry, physics, and 3D graphics.

The Scalar Triple Product

Given three vectors a , b , c , their scalar triple product (also called the box product or mixed product) is defined as:

Scalar triple product: $[a \ b \ c] = a \cdot (b \times c) = b \cdot (c \times a) = c \cdot (a \times b)$

The scalar triple product can be computed as a 3×3 determinant:

$$[a \ b \ c] = \det$$

a_1	a_2	a_3
b_1	b_2	b_3
c_1	c_2	c_3

Geometric Interpretation — Volume of a Parallelepiped

The absolute value of the scalar triple product equals the volume of the parallelepiped (a 3D solid with six parallelogram faces) formed by the three vectors a , b , c placed from the same origin:

Volume formula: Volume of parallelepiped $= |[a \ b \ c]| = |a \cdot (b \times c)|$

Area of a Parallelogram and Triangle

The cross product directly gives the area of a parallelogram:

Parallelogram area: Area of parallelogram formed by a and $b = |a \times b|$

Triangle area: Area of triangle formed by a and $b = (1/2) |a \times b|$

Coplanarity Condition

Three vectors a, b, c are coplanar (lie in the same plane) if and only if their scalar triple product is zero: $[a \ b \ c] = 0$. This is because a zero scalar triple product means the parallelepiped has zero volume — the three vectors lie in a flat plane.

Worked Example: Volume of Parallelepiped and Coplanarity

Given $a = \hat{i} + 2\hat{j} - \hat{k}$, $b = 2\hat{i} + \hat{j} + 3\hat{k}$, $c = -\hat{i} + \hat{j} + 2\hat{k}$.

- (i) Find the volume of the parallelepiped formed by a, b, c .
- (ii) Test whether $d = 3\hat{i} + 4\hat{j} + 2\hat{k}$, $e = \hat{i} + \hat{j} + \hat{k}$, $f = 2\hat{i} + 3\hat{j} + \hat{k}$ are coplanar.

Step 1: Set up the determinant for $[a \ b \ c]$:

$$[a \ b \ c] = \det$$

1	2	-1
2	1	3
-1	1	2

Step 2: Expand along Row 1:

$$= 1 \cdot \det([[1,3],[1,2]]) - 2 \cdot \det([[2,3],[-1,2]]) + (-1) \cdot \det([[2,1],[-1,1]])$$

$$= 1 \cdot (2-3) - 2 \cdot (4+3) - 1 \cdot (2+1)$$

$$= 1 \cdot (-1) - 2 \cdot (7) - 1 \cdot (3)$$

$$= -1 - 14 - 3 = -18$$

Step 3: Volume = $|[a \ b \ c]| = |-18| = 18$ cubic units.

Step 4: Test coplanarity of d, e, f:

$$[d \ e \ f] = \det$$

3	4	2
1	1	1
2	3	1

$$= 3(1-3) - 4(1-2) + 2(3-2) = 3(-2) - 4(-1) + 2(1) = -6 + 4 + 2 = 0$$

Since $[d \ e \ f] = 0$, the vectors d, e, f are coplanar. ✓

Worked Example: Area of Triangle Using Cross Product

Find the area of the triangle with vertices A(1,1,0), B(2,0,1), C(0,2,1).

Step 1: Compute edge vectors from A:

$$\vec{AB} = B - A = (2-1)\hat{i} + (0-1)\hat{j} + (1-0)\hat{k} = \hat{i} - \hat{j} + \hat{k}$$

$$\vec{AC} = C - A = (0-1)\hat{i} + (2-1)\hat{j} + (1-0)\hat{k} = -\hat{i} + \hat{j} + \hat{k}$$

Step 2: Compute $\vec{AB} \times \vec{AC}$:

$$\vec{AB} \times \vec{AC} = \det$$

$\hat{i}$	$\hat{j}$	$\hat{k}$
1	-1	1
-1	1	1

$$= \hat{i}(-1-1) - \hat{j}(1+1) + \hat{k}(1-1) = -2\hat{i} - 2\hat{j} + 0\hat{k}$$

Step 3: Magnitude: $|\vec{AB} \times \vec{AC}| = \sqrt{4 + 4 + 0} = \sqrt{8} = 2\sqrt{2}$

Step 4: Area of triangle = $(1/2)|\vec{AB} \times \vec{AC}| = (1/2)(2\sqrt{2}) = \sqrt{2}$ square units.

✓ Let's Check

Q.No	Type	Question
1	MCQ	The cross product $\vec{a} \times \vec{b}$ is a vector that is perpendicular to:
2	MCQ	If the scalar triple product $[\vec{a} \ \vec{b} \ \vec{c}] = 0$, the three vectors are:
3	Fill in Blank	The area of a parallelogram formed by vectors $\vec{a}$ and $\vec{b}$ equals $ \underline{\hspace{1cm}} $.
4	Fill in Blank	The cross product of two parallel vectors is the $\underline{\hspace{1cm}}$ vector.
5	True/False	$\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$ for any vectors $\vec{a}$ and $\vec{b}$.
6	True/False	The volume of a parallelepiped formed by vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ equals $ \vec{a} \cdot (\vec{b} \times \vec{c}) $.

Answers:

Q.No	Answer
1	c) Both a and b — the cross product is perpendicular to both input vectors (it is the normal to their plane).
2	b) Coplanar — zero scalar triple product means zero volume, so they lie in the same plane.
3	$a \times b$
4	zero
5	False — the cross product is anti-commutative: $a \times b = -(b \times a)$.
6	True — Volume = $ [a \ b \ c] = a \cdot (b \times c) $.

■ Case Study: Google PageRank and Eigenvalue Analysis — A Study in Applied Linear Algebra

Background:

When Sergey Brin and Larry Page founded Google in 1998, the core innovation was the PageRank algorithm — a method for ranking billions of web pages by importance. The mathematical foundation of PageRank is an eigenvalue problem on a massive matrix, directly applying concepts covered in this module. India's digital infrastructure — managed by NIXI (National Internet Exchange of India) and powering platforms like Flipkart, Swiggy, and Zomato — is deeply dependent on algorithms that originate in eigenvalue theory.

The Mathematical Model:

Consider the internet as a directed graph: each web page is a node, and each hyperlink is a directed edge. Construct an $n \times n$ matrix M (where n = number of web pages), called the link matrix or transition matrix, where $M_{ij} = 1/(\text{number of links on page } j)$ if page j links to page i , and 0 otherwise. Each column of M sums to 1 (it is a stochastic matrix).

The PageRank vector r — which gives the relative importance of every page — is the eigenvector corresponding to the dominant eigenvalue $\lambda = 1$ of this matrix: $M \cdot r = r$. The dominant eigenvector is the stable distribution of a random surfer who follows links at random.

Connection to This Module:

- **Dominant Eigenvalue:** The PageRank vector corresponds to the eigenvalue $\lambda = 1$. For any stochastic matrix, $\lambda = 1$ is always the dominant eigenvalue (all other eigenvalues have absolute value ≤ 1).
- **Power Method:** Google originally computed PageRank using the Power Method (precisely Rayleigh's Power Iteration). Starting from a uniform vector, repeated multiplication by M converges to the PageRank

eigenvector. For Google's first index of ~25 million pages, this required approximately 50–100 iterations to converge.

- **Eigenvalue Properties:** The fact that stochastic matrices always have dominant eigenvalue 1 (used in PageRank) is a consequence of the property that the eigenvalues of a matrix are bounded by its row/column norms — directly connected to the properties studied in Sub-unit 2.3.
- **Vector Normalisation:** After each power iteration, the vector is normalised (converted to unit vector) exactly as described in Sub-unit 2.4 — the Rayleigh quotient ($m_1, m_2, \dots$) converges to 1.

Scale and Impact:

Modern Google processes a matrix of approximately 1.7 trillion web pages. Computing the dominant eigenvector of such a matrix directly is impossible — but Power Method, operating on the sparse structure (most pages link to very few others), makes it feasible. The same mathematics powers recommendation systems at Netflix India, Hotstar, and Amazon India — their recommendation matrices are analysed for dominant eigenvectors to identify the "most influential" content preferences among millions of users.

Module Summary

Module 2 has covered two profound and deeply interconnected areas of mathematics — eigenvalue theory and vector algebra — each with vast applications in engineering, computing, physics, and data science.

In the eigenvalue section, the journey began with the characteristic equation $\det(A - \lambda I) = 0$, which systematically extracts the eigenvalues of any square matrix. Eigenvectors were then found by solving the homogeneous system $(A - \lambda I)X = 0$ for each eigenvalue. You learned the powerful properties that eigenvalues satisfy — the trace-sum relation, the determinant-product relation, and the Cayley-Hamilton theorem — which provide verification tools and computational shortcuts.

Rayleigh's Power Method demonstrated how the dominant eigenvalue can be found iteratively for large matrices — the algorithm behind Google's PageRank and modern recommendation engines. Diagonalization showed how a matrix can be expressed as $A = PDP^{-1}$, transforming complex operations like matrix powers into trivial ones. The Rank-Nullity theorem provided a fundamental bridge between the algebraic structure of a matrix and its geometric action as a linear transformation.

Key results from eigenvalue theory:

- Eigenvalues satisfy $\det(A - \lambda I) = 0$ (characteristic equation); the roots of this polynomial are the eigenvalues.
- Eigenvectors are found by solving $(A - \lambda I)X = 0$ for each λ ; they are non-zero and non-unique.
- Sum of eigenvalues = $\text{trace}(A)$; product of eigenvalues = $\det(A)$.
- Every matrix satisfies its own characteristic equation (Cayley-Hamilton).
- $A = PDP^{-1}$ when A has n linearly independent eigenvectors.

In the vector algebra section, vectors were introduced as directed quantities described by components in 3D space. Magnitude, direction cosines, and unit vectors provide

the tools for geometric description. The triangle and parallelogram laws govern vector addition.

Key results from vector algebra:

- Dot product: $a \cdot b = |a||b|\cos\theta$ — measures alignment; zero if perpendicular.
- Cross product: $|a \times b| = |a||b|\sin\theta$ — gives area of parallelogram; direction by right-hand rule.
- Scalar triple product: $[a \ b \ c] = a \cdot (b \times c)$ — gives volume of parallelepiped; zero if coplanar.
- Vector projection of a onto b : $[(a \cdot b)/(b \cdot b)] \cdot b$.
- Area of triangle with edges a, b : $(1/2)|a \times b|$.

Glossary

Term	Definition
Eigenvalue (λ)	A scalar associated with a square matrix A such that there exists a non-zero vector X with $AX = \lambda X$. Found by solving $\det(A - \lambda I) = 0$.
Eigenvector	A non-zero vector X satisfying $AX = \lambda X$ for some eigenvalue λ . Any scalar multiple of an eigenvector is also an eigenvector.
Characteristic Polynomial	The polynomial $\det(A - \lambda I)$, whose roots are the eigenvalues of A . For an $n \times n$ matrix, it has degree n .
Cayley-Hamilton Theorem	Every square matrix satisfies its own characteristic equation: if $p(\lambda) = 0$ is the characteristic equation, then $p(A) = O$.
Diagonalization	Writing a matrix $A = PDP^{-1}$, where D is diagonal (eigenvalues) and P is the modal matrix (eigenvectors as columns).
Rayleigh's Power Method	An iterative numerical method for finding the dominant eigenvalue by repeated multiplication and normalisation of a starting vector.
Magnitude of a Vector	The length $ a = \sqrt{a_1^2 + a_2^2 + a_3^2}$ — the scalar size of a vector, independent of direction.
Direction Cosines	The cosines of the angles a vector makes with the x, y, z axes: $(a_1/ a , a_2/ a , a_3/ a)$. Their squares sum to 1.
Dot Product	$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3 = a b \cos\theta$. A scalar measure of the alignment between two vectors.
Scalar Triple Product	$[a \ b \ c] = a \cdot (b \times c)$ — the volume of the parallelepiped formed by three vectors; zero if and only if the vectors are coplanar.

Self Assessment Questions

Section A: Multiple Choice Questions (MCQ)

Choose the most appropriate answer for each question.

1. The eigenvalues of a matrix are obtained by solving:
 - a) $AX = O$
 - b) $\det(A - \lambda I) = 0$
 - c) $\det(A) = \lambda$
 - d) $A - \lambda I = O$
2. If A has eigenvalues 2 and -3 , then $\det(A)$ equals:
 - a) -1
 - b) 5
 - c) -6
 - d) 6
3. If A has eigenvalues 2 and -3 , then $\text{trace}(A)$ equals:
 - a) -1
 - b) 5
 - c) -6
 - d) 6
4. Which statement about eigenvectors is correct?
 - a) Every matrix has at least one eigenvector
 - b) Eigenvectors are unique for each eigenvalue
 - c) Any non-zero scalar multiple of an eigenvector is also an eigenvector
 - d) The zero vector can be an eigenvector
5. The Cayley-Hamilton theorem states that a matrix satisfies:
 - a) Its own inverse equation

-
- b) Its own characteristic equation
- c) $AX = \lambda X$ for all X
- d) $A + A^T = 2A$
6. Rayleigh's Power Method is used to find the:
- a) Minimum eigenvalue
- b) Sum of all eigenvalues
- c) Dominant (largest magnitude) eigenvalue
- d) Inverse of the matrix
7. A matrix A is diagonalizable if it has:
- a) A zero determinant
- b) All identical eigenvalues
- c) n linearly independent eigenvectors (for $n \times n$ matrix)
- d) A symmetric structure
8. For the standard unit vectors: $\hat{i} \times \hat{j}$ equals:
- a) $\hat{i}$
- b) $-\hat{k}$
- c) $\hat{k}$
- d) $\hat{j}$
9. The magnitude of the vector $3\hat{i} - 4\hat{k}$ is:
- a) 7
- b) 1
- c) 5
- d) 25
10. Direction cosines of a vector satisfy:
- a) $\cos \alpha + \cos \beta + \cos \gamma = 1$
- b) $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

- c) $\cos \alpha \cdot \cos \beta \cdot \cos \gamma = 1$
- d) $\cos \alpha = \cos \beta = \cos \gamma$
11. The dot product of $a = \hat{i} + 2\hat{j} - 2\hat{k}$ and $b = 3\hat{i} - \hat{j} + 2\hat{k}$ is:
- a) 3
- b) -1
- c) -4
- d) 5
12. Two vectors are perpendicular if and only if their dot product is:
- a) 1
- b) Equal to product of magnitudes
- c) 0
- d) Positive
13. The scalar projection of a onto b is given by:
- a) $a \cdot b$
- b) $(a \cdot b) / |b|$
- c) $|a||b|\sin\theta$
- d) $a \times b / |b|$
14. The cross product $a \times b$ produces:
- a) A scalar equal to $|a||b|\cos\theta$
- b) A scalar equal to $|a||b|\sin\theta$
- c) A vector perpendicular to both a and b
- d) A vector parallel to a
15. The area of a parallelogram formed by vectors a and b equals:
- a) $a \cdot b$
- b) $|a \times b|$
- c) $(1/2)|a \times b|$

d) $|a||b|\cos\theta$

16. If $a \times b = 0$, it implies:

- a) $a = 0$ or $b = 0$
- b) a is perpendicular to b
- c) a and b are parallel (or one is zero)
- d) $a = b$

17. The scalar triple product $[a \ b \ c]$ gives:

- a) Area of triangle formed by a, b, c
- b) Volume of the parallelepiped formed by a, b, c
- c) The unit vector perpendicular to a, b, c
- d) The angle between a and $b \times c$

18. Three vectors a, b, c are coplanar if:

- a) $a \cdot b = 0$
- b) $a \times b = c$
- c) $[a \ b \ c] = 0$
- d) $|a| = |b| = |c|$

19. The rank-nullity theorem for an $m \times n$ matrix A states:

- a) $\text{rank}(A) + \text{nullity}(A) = m$
- b) $\text{rank}(A) \times \text{nullity}(A) = n$
- c) $\text{rank}(A) + \text{nullity}(A) = n$
- d) $\text{rank}(A) - \text{nullity}(A) = m$

20. In diagonalization $A = PDP^{-1}$, the diagonal matrix D contains:

- a) The eigenvectors of A
- b) The eigenvalues of A
- c) The cofactors of A
- d) The rows of A

Section B: True / False

State whether each statement is True or False.

1. A matrix with all distinct eigenvalues is always diagonalizable.
2. The cross product is commutative: $a \times b = b \times a$ for any vectors a and b .
3. The eigenvectors of a real symmetric matrix corresponding to different eigenvalues are always orthogonal.
4. The scalar triple product $[a \ b \ c]$ is positive when the three vectors form a right-handed system.
5. Rayleigh's Power Method converges to the smallest eigenvalue of a matrix.

Section C: Short Answer / Problem Solving

Answer each question with full working, including all intermediate steps.

1. Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. Verify your eigenvectors by substituting back into $AX = \lambda X$.
2. For the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, verify the Cayley-Hamilton theorem by finding the characteristic polynomial and showing $p(A) = O$.
3. Given vectors $a = 2\hat{i} + \hat{j} - 2\hat{k}$ and $b = \hat{i} + 3\hat{j} + \hat{k}$: (i) Find the angle between a and b . (ii) Find the projection of a onto b . (iii) Find $a \times b$ and verify it is perpendicular to both a and b .
4. Find the volume of the parallelepiped formed by the vectors $a = \hat{i} + 2\hat{j} + \hat{k}$, $b = 2\hat{i} - \hat{j} + \hat{k}$, and $c = \hat{i} + \hat{j} + 2\hat{k}$. Also find the area of the parallelogram formed by a and b .
5. Apply Rayleigh's Power Method to find the dominant eigenvalue of $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ using the initial vector $X_0 = [1; 1]$. Perform 3 iterations and comment on convergence towards the exact dominant eigenvalue.

Self Assessment — Answer Keys

Section A — MCQ

Q.No	Answer	Option Text
1	B	b) $\det(A - \lambda I) = 0$
2	C	c) -6
3	A	a) -1
4	C	c) Any non-zero scalar multiple of an eigenvector is also an eigenvector
5	B	b) Its own characteristic equation
6	C	c) Dominant (largest magnitude) eigenvalue
7	C	c) n linearly independent eigenvectors (for $n \times n$ matrix)
8	C	c) $\hat{k}$
9	C	c) 5
10	B	b) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
11	B	b) -1
12	C	c) 0
13	B	b) $(a \cdot b) / b $
14	C	c) A vector perpendicular to both a and b
15	B	b) $ a \times b $
16	C	c) a and b are parallel (or one is zero)
17	B	b) Volume of the parallelepiped formed by a, b, c

18	C	c) $[a \ b \ c] = 0$
19	C	c) $\text{rank}(A) + \text{nullity}(A) = n$
20	B	b) The eigenvalues of A

Section B — True / False Answer Key

Q.No	Answer	Notes
1	True	Correct.
2	False	Incorrect — see the relevant sub-unit for the accurate statement.
3	True	Correct.
4	True	Correct.
5	False	Incorrect — see the relevant sub-unit for the accurate statement.

Section C — Short Answer Model Solutions

Q.No	Model Solution
1	Characteristic equation: $\det(A - \lambda I) = (3 - \lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2) = 0$. Eigenvalues: $\lambda_1 = 4, \lambda_2 = 2$. For $\lambda_1 = 4$: $(A - 4I)X = 0 \rightarrow [-1, 1; 1, -1]X = 0 \rightarrow x_1 = x_2 \rightarrow X_1 = [1; 1]$. Verify: $A[1; 1] = [4; 4] = 4[1; 1] \checkmark$. For $\lambda_2 = 2$: $(A - 2I)X = 0 \rightarrow [1, 1; 1, 1]X = 0 \rightarrow x_1 = -x_2 \rightarrow X_2 = [1; -1]$. Verify: $A[1; -1] = [2; -2] = 2[1; -1] \checkmark$.
2	Characteristic polynomial: $p(\lambda) = \det([1 - \lambda, 2; 3, 4 - \lambda]) = (1 - \lambda)(4 - \lambda) - 6 = \lambda^2 - 5\lambda + 4 - 6 = \lambda^2 - 5\lambda - 2$. By Cayley-Hamilton: $p(A) = A^2 - 5A - 2I = O$.

	Compute $A^2 = [[1,2],[3,4]] \cdot [[1,2],[3,4]] = [[7,10],[15,22]]$. Check: $A^2 - 5A - 2I = [[7,10],[15,22]] - [[5,10],[15,20]] - [[2,0],[0,2]] = [[0,0],[0,0]] = O \checkmark$.
3	(i) $a \cdot b = (2)(1) + (1)(3) + (-2)(1) = 2 + 3 - 2 = 3$. $a = \sqrt{4+1+4} = 3$, $b = \sqrt{1+9+1} = \sqrt{11}$. $\cos \theta = 3/(3\sqrt{11}) = 1/\sqrt{11}$. $\theta = \cos^{-1}(1/\sqrt{11}) \approx 72.45^\circ$. (ii) $\text{proj}_b(a) = [(a \cdot b)/(b \cdot b)]b = [3/11](\hat{i} + 3\hat{j} + \hat{k}) = (3/11)\hat{i} + (9/11)\hat{j} + (3/11)\hat{k}$. (iii) $a \times b = \det[[\hat{i}, \hat{j}, \hat{k}], [2, 1, -2], [1, 3, 1]] = \hat{i}(1+6) - \hat{j}(2+2) + \hat{k}(6-1) = 7\hat{i} - 4\hat{j} + 5\hat{k}$. Verify: $(7)(2) + (-4)(1) + (5)(-2) = 14 - 4 - 10 = 0 \checkmark$ and $(7)(1) + (-4)(3) + (5)(1) = 7 - 12 + 5 = 0 \checkmark$.
4	Volume: $[a \ b \ c] = \det[[1,2,1],[2,-1,1],[1,1,2]] = 1(-2-1) - 2(4-1) + 1(2+1) = -3 - 6 + 3 = -6$. Volume = $-6 = 6$ cubic units. Area of parallelogram (a and b): $a \times b = \det[[\hat{i}, \hat{j}, \hat{k}], [1,2,1],[2,-1,1]] = \hat{i}(2+1) - \hat{j}(1-2) + \hat{k}(-1-4) = 3\hat{i} + \hat{j} - 5\hat{k}$. $a \times b = \sqrt{9+1+25} = \sqrt{35}$. Area = $\sqrt{35} \approx 5.92$ sq units.
5	Iteration 1: $Y_1 = [[4,1],[2,3]][1;1] = [5;5]$, $m_1 = 5$, $X_1 = [1;1]$. Iteration 2: $Y_2 = [5;5]$, $m_2 = 5$, $X_2 = [1;1]$. The vector $[1;1]$ is already an eigenvector of this matrix! Note: For $A = [[4,1],[2,3]]$, $\det = 10$, $\text{trace} = 7$. Eigenvalues: $\lambda^2 - 7\lambda + 10 = 0 \rightarrow$ $(\lambda - 5)(\lambda - 2) = 0 \rightarrow \lambda_1 = 5, \lambda_2 = 2$. Check: $A[1;1] = [5;5] = 5[1;1] \checkmark$. So power method converged in Iteration 1 since $X_0 = [1;1]$ happened to be the eigenvector for $\lambda = 5$ (dominant eigenvalue). m converged immediately to 5 $= \lambda_1 \checkmark$.

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